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Comparative Analyses of Age- and Sex-Specific Patterns of Vocal Behaviour in Four Species of Old World Monkeys

G. Hohmann

Indian Institute of Science, Bangalore, India; Max-Planck-Institut für Psychiatrie, München, FRG

Key Words. Vocal communication · Old World monkeys · *Macaca radiata* · *Macaca silenus* · *Presbytis johnii* · *Presbytis entellus* · Age and sex differences

Abstract. Sound recordings and behavioural data were collected from four primate species of two genera *(Macaca, Presbytis)*. Comparative analyses of structural and behavioural aspects of vocal communication revealed a high degree of intrageneric similarity but striking intergeneric differences. In the two macaque species *(Macaca silenus, Macaca radiata)*, males and females shared the major part of the repertoire. In contrast, in the two langurs *(Presbytis johnii, Presbytis entellus)*, many calls were exclusive to adult males. Striking differences between both species groups occurred with respect to age-specific patterns of vocal behaviour. The diversity of vocal behaviour was assessed from the number of different calls used and the proportion of each call in relation to total vocal output for a given age/sex class. In *Macaca,* diversity decreases with the age of the vocalizer, whereas in *Presbytis* the age of the vocalizer and the diversity of vocal behaviour are positively correlated. A comparison of the data of the two genera does not suggest any causal relationship between group composition (e.g. multi-male vs. one-male group) and communication system. Within each genus, interspecific differences in vocal behaviour can be explained by differences in social behaviour (e.g. group cohesion, intergroup relation, mating behaviour) and functional disparities. Possible factors responsible for the pronounced intergeneric differences in vocal behaviour between *Macaca* and *Presbytis* are discussed.

Introduction

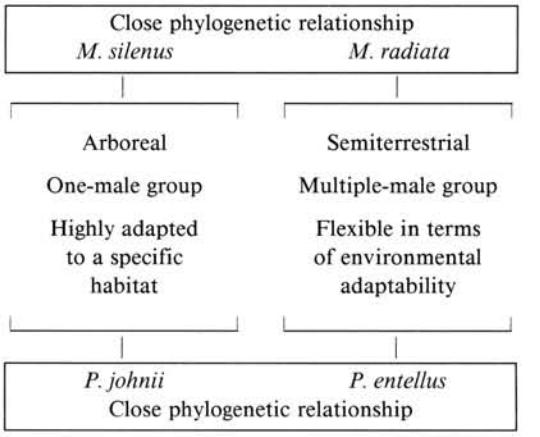
Most studies dealing with vocal communication in non-human primates have reported various forms of age- and sex-specific differences. In squirrel monkeys *(Saimiri sciureus)*, adult females exclusively utter a call ('chuck') during affiliative interactions with female group mates [1]. Nevertheless, the same call can be evoked from mature males by electrical brain stimulation [2]. Geissmann [3] reported that a female gibbon *(Hylobates pileatus)* transiently uttered the male-specific part of the territorial call. These and related findings [4–6] suggest that age- and sex-specific differences in the size and composition of the vocal repertoire within a given species may reflect differences

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of the quality of social interactions rather than the ability or inability to produce particular cells. In *Gorilla gorilla beringei*, silverback males vocalize more frequently and utilize a larger part of the repertoire than any other age/sex class [7]. Conversely, the vocal activity of the adult male of *Cercopithecus pogonias* is virtually confined to a single pattern [8]. It has been suggested that, apart from the genetic determination of species-specific behaviour, particular traits of vocal behaviour may be related to the social structure of the community [9]. Moreover, environmental factors may have a profound impact not only on the physical design of the signals but also on the quality and frequency of vocal interactions [10, 11].

The interrelations between structure, function and utilization of the different components of the vocal system, on the one hand, and social and environmental factors, on the other, can be investigated by comparing: (a) closely related species that differ in critical features, such as social organization and environmental adaptation, or (b) remotely related species in which social design and environmental adaptation are similar. Synopses of age- and sex-specific patterns of vocal behaviour were published by Eisenberg [12], Gautier and Gautier [13], and Snowdon [14]. However, because of technical and methodological disparities, most of these studies are of only limited value for further comparative analyses. A documentation of age- and sex-specific patterns of vocal behaviour in free-ranging chimpanzees *(Pan troglodytes)* and gorillas *(G. gorilla beringei)* has been presented by Marler and Tenaza [15]. However, whereas the data on chimpanzees were collected by the authors themselves, those on gorillas were derived from another study [7]. The chimpanzees belonged to a well-habituated, provisioned population, whereas information on gorillas came from poorly-habituated, unprovisioned groups. Recent observations of habituated gorilla groups reveal inaccuracies in previous reports [16].

To provide a basis for comparative analyses, a field project was launched to collect sound recordings and behavioural data from wild populations of four primate species living in the southern part of the Indian peninsula. This study involved two species of macaques *(Macaca silenus, Macaca radiata)* and two leaf monkeys *(Presbytis johnii, Presbytis entellus)*. The lion-tailed macaque *(M. silenus)* and the Nilgiri langur *(P. johnii)* are arboreal forest dwellers confined to restricted areas of the western Ghat mountain chain. The bonnet macaque *(M. radiata)* and the common langur *(P. entellus)* are semiterrestrial and adapted to a wide range of different environments. In contrast, the two forest-living species are typically organized in one-male groups (occasionally age-graded) with a comparatively low number of members [17, 18]. In order to assess correlations between the vocal system and social/environmental criteria, the following constellation was assumed:



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All four primate species mentioned above have been the subject of previous research [for reviews, see 19], but data on vocal communication were very fragmentary. Whereas structural and behavioural aspects of vocal communication of *M. silenus* have been analysed in a previous study [20], virtually nothing was known about vocal communication of *M. radiata*. Data on vocal communication of *P. entellus* from North India have been published by Vogel [11]. Poirier [21] provided an onomatopoetical description of the calls of *P. johnii* and specific aspects of vocal communication have been studied by Horwich [22] and Herzog and Hohmann [23].

To achieve a data set suitable for comparative analyses, it was necessary to collect adequate samples of data on vocalizations and behaviour for all four species, using the same equipment, sampling technical and observation methods.

Table 1. Size and composition of the focal groups of *M. silenus* (MS), *M. radiata* (MR), *P. johnii* (PJ) and *P. entellus* (PE) at the beginning of fieldwork

Study group	A♂	A♀	S♂	S♀	J♂	J♀	I ♂		I ♀	
							born 1986	born 1987	born 1986	born 1987
MS-1	1	6	1	1	2	3	2	1	–	3
MS-2	2	5	1	2	3	3	1	–	2	3
MR-1	6	11	2	2	12	10	3	7	6	4
MR-2	5	7	3	2	8	4	3	3	4	3
PJ-1	1	4	–	–	–	1	–	2	–	2
PJ-2	1	4	2	1	3	1	1	–	1	1
PE-1	3	9	2	6	8	8	?	5	3	4
PE-2	3	7	–	2	5	5	?	?	4	2

A = Adult; S = subadult; J = juvenile; I = infant.

Table 2. Age classification used in this study (figures indicate age in years)

Age class		*M. silenus*	*M. radiata*	*P. johnii*	*P. entellus*
Infant	♂ + ♀	0–1	0–1	0–1	0–1
Juvenile	♂	1–4	1–4	1–4	1–4
	♀	1–3	1–3	1–3	1–3
Subadult	♂	4–8	4–6	4–7	4–7
	♀	3–5	3–5	3–4	3–4
Adult	♂	≥ 8	≥ 6	≥ 7	≥ 7
	♀	≥ 5	≥ 5	≥ 4	≥ 4

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Methods

Fieldwork was carried out from December 1985 to November 1987 in forest areas in the southern part of the western Ghats (India). The two main study areas were located in the Anaimalay and Ashambu hill range (Tamil Nadu).

Behavioural data and sound recordings were gathered from habituated study groups (table 1). For only one species *(M. radiata)* did habituation require initial provisioning. Data collection commenced when the appearance of the observer no longer caused any obvious change in group activity. By that stage, most group members could be identified individually, and the observation distance varied between 2 and 20 m. The assessment of the age of animals not born during the course of fieldwork was based on morphology (e.g. body size, pigmentation, dentition) and behaviour (table 2).

Table 3. Size, composition and utilization of the vocal repertoire: *M. silenus* and *M. radiata*

Vocal pattern	*M. silenus*				*M. radiata*				Mode of equivalence
	A♂	A♀	J♂♀	I♂♀	A♂	A♀	J♂♀	I♂♀	
Contact whoo	x	x	x		x	x	x		C + S
Isolation whoo			x	x			x	x	C + S
Long-distance whoo	x	x	x		x	x	x		C + S
Greeting		x	x	x	x	x	x	x	C + S
Alarm call	x	x	x		x	x	x	?	C
Grunt/growl	x	x	x		x	x	x		C + S
Bark	x	x	x		x	x			C + S
Threat rattle	x	x	x	x	x	x	x	x	C + S
Female copulation call		x				x			C + S
Male copulation call	x				x				C + S
Squeak		x	x	x	x	x	x		C + ?
Scream		x	x	x	x	x	x	x	C + S
Shriek/squeal		x	x		x	x	x	x	C + S
Twit/tonal scream		x	x	x		x	x	x	C + S
Sneeze				x				x	C + S
Yap		x	x	x			x	x	C + S
Babble				x				x	C + S
Weeping			x	x	–	–	–	–	– –
Loud call	x				–	–	–	–	– –
Teeth grinding	x				–	–	–	–	– –
Milk grumble				x				?	– –
Mounting grunt	–	–	–	–	x				– –
Whistle	–	–	–	–		x	x	x	– –
Cluck	–	–	–	–			x	x	– –
Contact rattle	–	–	–	–	x	x	x	x	– –
Vibrato growl	–	–	–	–	x	x	x		– –
Chuckle	–	–	–	–	x	x	x		– –
Arched scream	–	–	–	–	x	x	x		– –
Hack	–	–	–	–		x	x	x	– –

C = Context equivalent; S = structure equivalent; A = adult; J = juvenile; I = infant.

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For sound recordings, a Uher-Monitor tape recorder (speed: 19 cm/s) and a Sennheiser directional microphone K3N (module M88, dynamic range 0.05–15 kHz) were used. In areas with high levels of background noise, a module ME40 (dynamic range 0.05–15 kHz) mounted to a parabolic reflector (focal length 165 mm) was used. Spectrographic analyses were made on a Kay Sonagraph 7800 and a FFT real-time analyser (Uniscan-II).

Observed context of emission
feeding, foraging, group movement
loss of physical contact, group fission
group fission, adult female after loss of infant
affiliative contact such as grooming/huddling
appearance of a potential predator
agonistic intra- and intergroup encounter
agonistic interactions within the group
general social tension, distance threat
copulation
copulation, precedes female vocalizations
agonistic behaviour without physical contact
agonistic interactions with physical contact
subordinates in response to severe attacks
harassment of subordinates, weaning
unspecific arousal, loss of contact, weaning
play, mild threat, chasing
unspecific arousal, mild maltreatment
physical and/or social isolation, weaning
group fission, loud calls of distant groups
threat directed against other males
nursing
dominance mounting between mature males
food competition, intergroup encounter
food competition, agonistic interaction
approach, marked change in group activity
distance threat, intergroup encounter
threat/harassment against subordinates
agonistic intra- and intergroup encounter
response to threat, maltreatment, weaning

The classification of the vocal repertoire was based on standard measurements of the physical parameters time, frequency and energy distribution. According to its physical features, each sample was placed into a particular sound class separated from other classes by a more or less distinct structural design. Classes connected by calls with intermediate/intergrading features were combined in categories. For further division, these categories were subjected to cluster analyses ('hill climbing'), taking into account additional physical data and context-relevant parameters. Detailed descriptions of the vocal repertoires of the four species and the procedures used for classification have been published in previous reports [20, 23–26].

For data collection, a given group was followed for 3–5 consecutive days. The time spent in acoustic and visual contact with the group varied with the stage of habituation and weather conditions (range: 4–14 h/day). During observation sessions, sounds were recorded ad libitum with the recording equipment set in permanent operation. In addition to the recordings of vocalizations, the age/sex of the vocalizer and data of the social interactions were recorded on the second track of the tape.

Initially most of the records were obtained by group scans (GS). As soon as a reliable identification of the different group members was possible, data were also collected on the basis of focal animal sampling (FS). The total times spent in contact with at least one member of one of the study groups was: *M. silenus*, 1,020 h (GS = 855; FS = 165); *M. radiata*, 912 h (GS = 800; FS = 112); *P. johnii*, 1,200 h (GS = 1,050; FS = 150); *P. entellus*, 700 h (GS = 600; FS = 100).

Assessments of the frequency of use of the different vocal patterns derived from group scans. Adequate samples of tape recordings of each species were systematically scanned, using a FFT real-time analyser with a flexible frequency-time display. By visual inspection and simultaneous acoustic control (via headphones), reliable identification of the different vocal patterns was made possible. Calls which could not be classified because of simultaneous vocalizations or for other reasons were disregarded. Vocalizations regularly emitted as phrases or bouts (e.g. loud call of *M. silenus* and *P. johnii*) were scored as single vocal event. The tapes selected for this procedure were recorded in situations of relaxed social and motor activity (mixed bouts of foraging and feeding; rest

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with play and grooming). In these situations, group cohesion was relatively strong, and conditions for sampling a larger proportion of the momentary vocal output of the group were reasonably good. Because of marked variation in the number of subjects within recording range, factors such as size and composition of the group were disregarded.

Analyses of the frequency of use of the different calls by the different age/sex classes are based on observations of focal animals. Under the given conditions, it was found that direct sampling by auditory discrimination of the main acoustic categories with prepared observation sheets was the most efficient method. All vocal events of a focal animal were tabulated into 5-min sampling periods (5 min = 1 epoch). In order to avoid bias due to interspecific differences in overall vocal activity, only such epochs with at least one vocal event (= active epochs), regardless of whether emitted by the focal animal or another group member, were considered. The total repertoire was divided into major categories with calls of similar acoustic features (e.g. clear calls of macaques, tonal frequency-modulated contact calls of langurs) noted under the same headings. Because of inadequate sampling, alarm calls and all non-vocal utterances were rejected from this analysis. The fact that infants of both langur species were frequently transferred from the mother to other female group mates occasionally

Table 4. Size, composition and utilization of the vocal repertoire: *P. johnii* and *P. entellus*

Vocal pattern	*P. johnii*				*P. entellus*				Mode of equivalence
	A♂	A♀	J♂♀	I♂♀	A♂	A♀	J♂♀	I♂♀	
Loud call	x				x				C + S
Harsh bark	x				x				C + S
Teeth grinding	x				x				C + S
Cough/grunt bark	x				x	x			C
Grunt	x	x			x	x			S
Rumble	x	x			x	x			C + S
Cough		x	x	x		x	x	x	C + S
Alarm call		x	x			x	x		C + S
Warble		x	x	x		x	x	x	C + S
Whistle		x	x	x		x	x	x	C + S
Shriek		x	x	x		x	x	x	C + S
Squeal		x	x	x		x	x	x	C + S
Hiccup	x	x	x	x	x	x	x	x	? + S
Wailing			x	x			x	x	C + S
Contact tremolo		x	x	?		x	x	?	C + S
Snarl		x			–	–	–	–	– –
Ho-ho call	x	x	x		–	–	–	–	– –
Pant bark	x				–	–	–	–	– –
Rumble scream	–	–	–	–	x				– –
Soft bark	–	–	–	–	x				– –
Honk	–	–	–	–	x				– –
Teeth chattering	–	–	–	–		x			– –
Isolation peep	–	–	–	–			x	x	– –
Milk grumble				?				x	– –

C = Context equivalent; S = structure equivalent; A = adult; J = juvenile; I = infant.

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led to misinterpretations concerning the infant's identity. Therefore, in this part of the study male and female infants were assigned to one class.

Results

Size and Composition of the Vocal Repertoire

In each species group *(Macaca, Presbytis)*, it was possible to divide the repertoires into one section containing co-called equivalent calls and another consisting of elements not found in the repertoire of the congeneric species. Calls of distinct physical structure uttered in identical or comparable situations (e.g. alarm calls evoked by predators, calls uttered during mating activities) were labelled context equivalent; calls of similar spectral and temporal features were named structure equivalent.

As shown in table 3 and 4, the repertoires of each species group included many equivalent vocal patterns. In the case of *Macaca,* however, a large number of calls recorded from bonnet macaques lacked equivalent elements in the repertoire of lion-tailed macaques, whereas in *Presbytis* the number of non-equivalent elements was relatively balanced between the two species. In the two macaque species, adult males and females shared almost the entire adult repertoire; 3 *(M. silenus)* and 2 *(M. radiata)* calls were uttered exclusively by male and female infants.

In the case of the two langur species, the data in table 4 indicate pronounced asymmetries between mature males and females. In *P. entellus*, 6 calls (29% of the total repertoire) were heard exclusively from mature males, and in *P. johnii* this applied to 5 calls (28% of the total repertoire). In each of the four species, 1 sound was given exclusively by mature females.

Evaluation of the Relative Frequencies of Signal Use

Table 5 shows combined data for two observation groups of each of the macaque species. In the case of *M. silenus*, more than 60% of the calls recorded were contact whoos; threat rattles ranked second (5.9%) and were followed by isolation whoos (5.4%) and alarm calls (5%). The remaining 17 vo-

Observed context of emission
group encounter, loud calls of other groups
appearance of predator (loud-call display)[1]
agonistic intra- and intergroup encounter
group encounter, group movement
affiliative approach[1], agonistic interaction[2]
playful mounting, embracing, huddling
mild agonism, general social tension
appearance of predator in close proximity
play, agonistic intra- and intergroup encounter
loss of contact, response to loud-call display
sexual harassment, response to loud calls
response to rejection of physical contact
appearance of predator[1], intergroup encounter[2]
loss of physical/visual contact, weaning
affiliative behaviour, huddling, embracing
response to approach/attack of adult male
part of the loud-call display
agonistic group encounter, loud-call display
agonistic intragroup interactions
response to a low-flying raptor
mild agonistic intragroup interactions
social grooming
loss of physical/visual contact
nursing

[1] Only *P. johnii*; [2] only *P. entellus.*

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cal patterns contributed approximately 20% to the total output.

In *M. radiata*, the use of the different calls was more balanced compared to the other macaque species. Contact rattles were the most frequently recorded calls (29.3% of total output). Threat rattles ranked second (13.8%), growls third (12%) and isolation whoos fourth (6.3%). The contact whoo, the most dominant vocalization of the other macaque species, ranked fifth in *M. radiata* (5.9%).

Table 5. Relative frequency of emission of the different vocal patterns by *M. silenus* and *M. radiata*

Vocal pattern	*M. silenus* (n = 7,133.5) (A)		*M. radiata* (n = 10,301.0) (B)		$FI = \frac{\lvert A-B \rvert}{A+B}$
Contact whoo	4,940.0	(69.25)	613.3	(5.95)	0.842
Threat rattle	420.9	(5.90)	1,312.1	(13.81)	0.401
Isolation whoo	391.8	(5.49)	650.6	(6.32)	0.070
Alarm call	365.3	(5.12)	545.2	(5.30)	0.017
Twit	233.8	(3.28)	341.1	(3.31)	0.005
Sneeze	134.2	(1.88)	143.6	(1.40)	0.146
Shriek	110.8	(1.55)	287.1	(2.79)	0.286
Greeting	109.5	(1.54)	165.0	(1.61)	0.022
Squeak	104.1	(1.46)	194.2	(1.89)	0.128
Growl	96.9	(1.36)	1,236.5	(12.00)	0.796
Bark	38.6	(0.54)	126.1	(1.22)	0.386
Yap	26.9	(0.38)	25.7	(0.25)	0.206
Babble	17.5	(0.26)	27.4	(0.27)	0.019
Non-tonal scream	16.2	(0.23)	564.3	(5.48)	0.919
Long-distance whoo	7.6	(0.10)	63.9	(0.62)	0.722
Female copulation call	4.5	(0.06)	0.8	(0.01)	0.714
Male copulation call	1.3	(0.02)	7.5	(0.07)	0.556
Weeping	96.9	(1.36)	–	–	
Milk grumble	8.5	(0.12)	–	–	
Loud call	4.9	(0.07)	–	–	
Teeth grinding	0.9	(0.01)	–	–	
Contact rattle	–	–	3,140.2	(30.48)	
Chuckle	–	–	269.7	(2.62)	
Whistle	–	–	195.0	(1.89)	
Arched scream	–	–	192.5	(1.87)	
Hack	–	–	131.1	(1.27)	
Vibrato growl	–	–	29.9	(0.29)	
Mounting grunt	–	–	26.6	(0.26)	
Cluck	–	–	11.6	(0.11)	

The first value (left) in each of the two main columns shows the real value normalized for a 10-hour time interval, followed (in parentheses) by the corresponding percent value representing the co-efficients A and B for the calculation of the index of frequency FI.

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To permit a direct comparison of the use of the equivalent vocal patterns, an index of frequency (FI) was introduced based on the percentage values shown in the left and centre columns of table 5, ranging between 0 (equality of usage) and 1 (maximum difference in use of a particular call). The index values for the two macaque species are shown in the right column of table 5. Pronounced differences in usage (FI ≥ 0.5) occurred with 6 of the 17 equivalent pairs. For 5 calls, a low index value (FI ≤ 0.1) indicated an almost identical use of equivalent basic patterns.

Corresponding data for the two langur species are shown in table 6. The call most frequently uttered by *P. johnii* was the whistle (18%), which ranked second for the other species; it was followed by the hiccup (16.5%) and the warble (13.5%), ranking eleventh and fourth for the other species.

The index values show that 4 out of the 15 equivalent pairs of calls were uttered with different frequencies by the two langur spe-

Table 6. Relative frequency of emission of the different vocal patterns of *P. johnii* and *P. entellus*

Vocal pattern	*P. johnii* (n = 8,288.9) (A)		*P. entellus* (n = 10,602.8) (B)		$FI = \frac{\lvert A-B \rvert}{A+B}$
Whistle	524.8	(18.35)	2,040.3	(19.24)	0.024
Hiccup	471.7	(16.49)	172.0	(1.62)	0.821
Warble	384.0	(13.43)	989.5	(9.33)	0.180
Squeal	326.7	(11.43)	1,478.8	(13.95)	0.099
Cough	308.5	(10.79)	348.7	(3.29)	0.533
Shriek	133.1	(4.65)	442.9	(4.18)	0.053
Teeth grinding	127.3	(4.45)	967.5	(9.13)	0.345
Cough/grunt bark	115.2	(4.03)	2,509.9	(23.76)	0.709
Contact tremolo	112.8	(3.95)	163.4	(1.54)	0.439
Alarm call[1]	90.7	(3.17)	69.1	(0.65)	0.660
Wailing	42.1	(1.47)	238.0	(2.24)	0.208
Grunt	38.3	(1.34)	256.8	(2.42)	0.287
Rumble	36.2	(1.27)	95.8	(0.90)	0.171
Harsh bark	29.7	(1.04)	193.2	(1.82)	0.273
Loud call	28.0	(0.98)	145.3	(1.37)	0.166
Ho-ho call	71.1	(2.49)	–	–	
Pant bark	16.2	(0.57)	–	–	
Snarl	3.5	(0.12)	–	–	
Isolation peep	–	–	284.3	(2.68)	
Rumble scream	–	–	53.4	(0.50)	
Milk grumble	–	–	40.1	(0.38)	
Honk	–	–	8.6	(0.08)	
Soft grunt	–	–	5.5	(0.05)	

Conventions as in table 5.

[1] Sample size (number of encounters with predators): *P. entellus*, n = 3/13 h; *P. johnii*, n = 7/29 h.

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Table 7. Age- and sex-specific differences of vocal behaviour in *M. silenus* as obtained by focal animal sampling

	A♂ (250)	A♀ (568)	S♂ (106)	S♀ (161)	J♂ (100)	J♀ (146)	I♂♀ (278)	Σ
1 Whoo call	101.8	783.8	189.1	401.7	106.8	286.0	953.4	2,822.6
	57.275	96.396	93.296	97.453	77.899	77.297	72.574	82.375
	3.607	27.770	6.699	14.232	3.784	10.133	33.777	100
2 Threat rattle + growl	69.6	11.8	6.8	5.2	2.4	19.7	3.2	118.7
	39.027	1.446	3.355	1.278	1.961	5.448	0.239	3.463
	58.640	9.941	5.753	4.380	2.022	16.596	2.696	100
3 Non-tonal scream	–	0.6	–	2.5	13.2	45.2	44.8	106.3
	–	0.074	–	0.607	9.628	12.216	3.410	3.102
	–	0.564	–	2.352	12.418	42.521	42.145	100
4 Greeting	–	7.1	6.8	2.0	1.5	6.6	5.5	29.5
	–	0.873	3.355	0.485	1.094	1.784	0.419	0.861
	–	24.068	23.051	6.780	5.085	22.373	18.644	100
5 Twit	–	–	–	–	0.5	1.6	1.7	3.8
	–	–	–	–	0.365	0.432	0.129	0.111
	–	–	–	–	13.158	42.105	44.737	100
6 Shriek/ squeal/ squeak	–	0.2	–	–	11.8	9.2	269.6	290.8
	–	0.025		–	8.607	2.486	20.522	8.487
	–	0.069	–	–	4.058	3.164	92.710	100
7 Yap	–	0.4	–	0.2	0.9	1.7	12.1	15.3
	–	0.049	–	0.049	0.656	0.459	0.921	0.447
	–	2.614	–	1.307	5.882	11.111	79.085	100
8 Female copulation call	–	7.3	–	–	–	–	–	7.3
	–	0.898	–	–	–	–	–	0.213
	–	100	–	–	–	–	–	100
9 Male copulation call	0.5	–	–	–	–	–	–	0.5
	0.303	–	–	–	-	–	–	0.016
	100	–	–	–	–	–	–	100
10 Bark	–	1.9	–	0.6	–	–	–	2.5
	–	0.234	–	0.146	–	–	-	0.073
	–	76.000	–	24.000	–	–	–	100
11 Sneeze	–	–	–	–	–	–	23.4	23.4
	–	–	–	–	–	–	1.781	0.683
	–	–	–	–	–	–	100	100
12 Loud call	5.8	–	–	–	–	–	–	5.8
	3.263	–	–	–	–	–	–	0.169
	100	–	–	–	–	–	–	100
Σ_{1-12}	177.7	813.1	202.7	412.2	137.1	370.0	1,313.4	3,426.5
	100	100	100	100	100	100	100	100
	5.187	23.729	5.916	12.030	4.001	10.798	38.339	100

A = Adult; S = subadult; j = juvenile; I = infant. The three figures in each column, reading from top to bottom, indicate (a) the normalized (10 h) real value of frequency of use calculated out of the total of active epochs (in parentheses), (b) the corresponding percent value with which a given call/category contributes to the total output of a given age/sex class and (c) the percentage with which each age/sex class contributes to the total output of the respective call/category.

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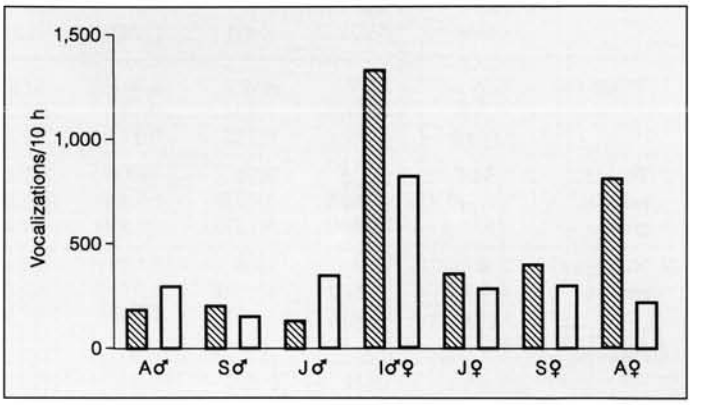


Fig. 1. Total vocal rates of the different age/sex classes obtained with focal animal sampling for macaques. Because of the differences in observation time of each age/sex class, values given below are normalized for an equivalent time interval of 10 h. ▨ = *M. silenus* (n 3,430 calls/10 h); □ = *M. radiata* (n = 2,430 calls/10 h); A = adult; S = subadult; J = juvenile; I = infant.

cies (FI ≥ 0.5), whereas for 3 calls the FI was below 0.1, indicating strong symmetry in usage.

Patterns of Age- and Sex-Specificity of Vocal Behaviour

The data shown in table 3 and 4 indicate varying degrees of age and sex specificity in the use of the vocal repertoires for the four species. For a more detailed analysis of these differences in vocal behaviour, data from focal animals were collected in the field. Except for subadult males of *P. johnii* and *M. silenus*, focal animal analyses were based on samples of at least three different individuals of each age/sex class.

Figure 1 illustrates overall vocal activity for the seven age/sex classes of the two macaque species. In the case of *M. silenus*, females of all age classes vocalized more often than males; the highest vocal rate was recorded for adult females, and juvenile males accounted for the lowest rate. Similar differences in the overall vocal activity between different age/sex classes also occurred in *M. radiata*. However, interclass differences were less pronounced than in *M. silenus* and, in terms of sex specificity, less coherent.

Frequencies of use of the different vocal categories by the seven age/sex classes of *M. silenus* are presented in table 7. Pronounced differences existed in the overall use of different sound categories by the various age/sex classes, ranging from 8 (adult females and infants) to 3 (subadult males). Moreover, it can be seen that whoo calls dominated all other vocalizations in all classes. Within each age class, females uttered consistently more whoo calls than their male peers. Among females, there was a positive correlation between rate of whoo calling and age of vocalizer. A converse trend seems to exist in the second category (threat rattle/growl) where males (except juveniles) were more active than their female peers. In all other categories, age and sex specificity was less marked.

In *M. radiata* (table 8) infants using 10 different categories accounted for the highest number whereas subadult males used only 6 different categories. Neglecting sex-specific differences, the rates of the second category, including the threat rattle and growl, increased with the age of the vocalizer. In the case of contact rattles and greetings, however, age of vocalizer and rate of vocaliza-

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Table 8. Age- and sex-specific differences of vocal behaviour in *M. radiata* as obtained by focal animal sampling

	A♂ (130)	A♀ (142)	S♂ (142)	S♀ (124)	J♂ (159)	J♀ (119)	I♂♀ (264)	Σ
1 Whoo call	0.9	22.8	4.2	2.9	81.5	74.3	374.8	561.4
	0.306	10.200	2.824	0.957	23.272	19.866	50.876	23.100
	0.160	4.061	0.748	0.517	14.517	13.235	66.762	100
2 Threat rattle + growl	264.7	111.5	80.8	197.6	92.1	99.4	38.8	884.9
	89.911	49.883	54.338	65.240	26.299	26.578	5.267	36.410
	29.913	12.600	9.131	22.330	10.408	11.233	4.385	100
3 Non-tonal scream	2.8	61.7	18.6	53.2	67.2	23.4	84.3	311.2
	0.951	27.604	12.508	17.563	19.189	6.257	11.443	12.804
	0.900	19.826	5.977	17.095	21.594	7.519	27.089	100
4 Greeting	–	4.2	3.4	1.0	0.8	2.0	2.1	13.5
	–	1.880	2.286	0.330	0.228	0.535	0.285	0.600
	–	31.111	25.185	7.407	5.926	14.814	15.600	100
5 Twit	–	–	1.7	7.7	13.6	5.3	13.5	41.8
	–	–	1.143	2.542	3.883	1.417	1.832	1.720
	–	–	4.067	18.421	32.536	12.679	32.297	100
6 Shriek squeal squeak	–	–	–	1.0	4.5	3.2	45.5	54.2
	–	–	–	0.330	1.285	0.856	6.176	2.230
	–	–	–	1.845	8.303	5.904	83.948	100
7 Yap	–	2.5	–	–	0.8	–	9.5	12.8
	–	1.118	–	–	0.228	–	1.290	0.530
	–	19.531	–	–	6.250	–	74.219	100
8 Female copulation call	–	0.5	–	–	–	–	–	0.5
	–	0.233	–	–	–	–	–	0.021
	–	100	–	–	–	–	–	100
9 Male copulation call	7.4	–	–	–	–	–	–	7.4
	2.513	–	–	–	–	–	–	0.305
	100	–	–	–	–	–	–	100
10 Bark	0.9	–	–	–	–	–	–	0.9
	0.306	–	–	–	–	–	–	0.037
	100	–	–	–	–	–	–	100
11 Sneeze	–	–	–	–	–	–	60.7	60.7
	–	–	–	–	–	–	8.239	2.500
	–	–	–	–	–	–	100	100
12 Contact rattle	17.7	20.3	40.0	39.5	73.9	73.7	82.7	347.8
	6.012	9.082	26.900	13.011	21.102	19.706	11.226	14.310
	5.089	5.837	11.501	11.357	21.248	21.190	23.778	100
13 Whistle	–	–	–	–	15.8	92.7	24.8	133.3
	–	–	–	–	4.512	24.786	3.366	5.485
	–	–	–	–	11.853	69.542	18.604	100
Σ_{1-13}	294.4	223.5	148.7	302.9	350.2	374.0	736.7	2,430.4
	100	100	100	100	100	100	100	100
	12.113	9.197	6.120	12.463	14.410	15.390	30.312	100

Conventions as in table 7.

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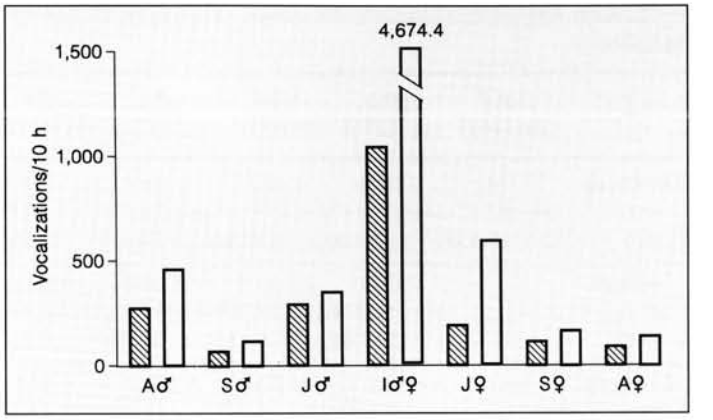


Fig. 2. Total vocal rate of the different age/sex classes obtained with focal animal observations for langurs. Because of the differences in observation time of each age/sex class, values given below are normalized for an equivalent time interval of 10 h. ▨ = *P. johnii* (n = 1,980 calls/10 h); □ = *P. entellus* (n = 6,400 calls/10 h); A = adult; S = subadult; J = juvenile; I = infant.

tion showed an inverse correlation. The most obvious sex differences occurred in adults, where – except for one category (threat rattle/growl) – the vocal activity of females was consistently higher compared to that of males.

Figure 2 shows the overall vocal rates for the two langur species. During equivalent time intervals (10 h) the total vocal output of *P. entellus* exceeded 6,000 calls, while less than 2,000 calls were recorded for the other species. Apart from these general differences, the data for the different age/sex classes were rather similar. In both species, infants uttered the highest number of calls, whereas subadult males showed the lowest vocal activity.

The utilization of 10 vocal categories by the different age/sex classes of *P. johnii* is detailed in table 9. The number of vocal categories used by the different age/sex classes ranged from 5 (infants) to 8 (juvenile males). In the case of the hiccup, the only pattern uttered by all of the seven age/sex classes, females were more active than males, and the rate of emission increased with the age of the vocalizer. Although less consistent, a similar trend of age-related activity seems to exist in the case of coughs, whereas the utterance of modulated contact calls (warble, whistle) decreased with the age of the vocalizer.

In *P. entellus* (table 10), age and sex differences in hiccup and cough vocalizations were less consistent. The number of shrieks and modulated contact calls decreased with the age of the vocalizer. Infants, although dominant in terms of overall vocal activity, used only 5 different categories, while the repertoire of adult females, which produced the second lowest vocal rate, consisted of 10 different vocal categories.

The figures shown in tables 5 and 10 were converted to index values, following a procedure introduced by Shannon and Weaver [27]. This involves the calculation of two indices, $H_i max$ and H_i. $H_i max$ represents the value if a given vocal pattern is uttered by all age/sex classes with the same frequency, while H_i represents the actual observed value. The formulae used for this calculation are:

$$H_i = -\sum_{i=1}^{n} P(i)_i \log_2 [P(i)] \text{ and}$$

$$H_i max = \log_2 a,$$

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Table 9. Age- and sex-specific differences of vocal behaviour in *P. johnii* as obtained by focal animal sampling

	A♂ (162)	A♀ (175)	S♂ (74)	S♀ (131)	J♂ (135)	J♀ (114)	I♂♀ (250)	Σ
1 Hiccup	14.1	19.9	3.2	25.6	8.9	11.6	5.2	88.5
	5.242	28.147	5.818	25.447	3.168	6.717	0.504	4.469
	15.932	22.486	3.616	28.927	10.056	13.107	5.876	100
2 Cough		37.0	35.7	36.6	8.0	6.3	13.0	136.6
		52.334	64.909	36.382	2.848	3.648	1.261	6.898
		27.086	26.135	26.794	5.857	4.612	9.517	100
3 Warble	–	2.1	8.1	18.3	211.6	142.1	926.5	1,308.7
+	–	2.970	14.727	18.191	75.329	82.281	89.838	66.089
whistle	–	0.160	0.619	1.398	16.169	10.858	70.795	100
4 Contact	–	6.2	–	10.1	–	2.1	–	18.4
tremolo	–	8.769	–	10.040	–	1.216	–	0.929
	–	33.696	–	54.891	–	11.413	–	100
5 Shriek	–	–	–	0.9	44.4	2.1	68.2	115.6
	–	–	–	0.895	15.806	1.216	6.613	5.838
	–	–	–	0.779	38.408	1.817	58.997	100
6 Squeal	–	3.4	–	1.8	5.3	5.3	18.4	34.2
	–	4.809	–	1.789	1.887	3.069	1.784	1.727
	–	9.942	–	5.263	15.497	15.497	53.801	100
7 Grunt	34.8	1.4	3.2	–	0.9	–	–	40.3
	12.937	1.980	5.818	–	0.320	–	–	2.035
	86.352	3.474	7.940	–	2.233	–	–	100
8 Loud call	8.9	–	1.6	–	–	–	–	10.5
	3.309	–	2.909	–	–	–	–	0.529
	84.762	–	15.238	–	–	–	–	100
9 Cough bark	203.0	–	3.2	–	0.9	–	–	207.1
	75.464	–	5.818	–	0.320	–	–	10.459
	98.020	–	1.560	–	0.435	–	–	100
10 Ho-ho call	8.2	0.7	–	7.3	0.9	3.2	–	20.3
	3.048	0.990	–	7.256	0.320	1.853	–	1.024
	40.394	3.448	–	35.961	4.433	15.764	–	100
Σ_{1-10}	269.0	70.7	55.0	100.6	280.9	172.7	1,031.3	1,980.2
	100	100	100	100	100	100	100	100
	13.584	3.570	2.777	5.080	14.185	8.721	52.081	100

Conventions as in table 7.

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Table 10. Age- and sex-specific differences of vocal behaviour in *P. entellus* as obtained by focal animal sampling

	A♂ (132)	A♀ (247)	S♂ (121)	S♀ (124)	J♂ (122)	J♀ (137)	I♂♀ (269)	Σ
1 Hiccup	6.4	9.2	–	5.8	7.9	6.1	2.6	38.0
	1.419	7.221	–	3.864	2.340	1.046	0.056	0.591
	16.842	24.211	–	15.263	20.789	16.053	6.842	100
2 Cough	2.7	17.0	1.0	1.0	13.8	0.9	–	36.4
	0.599	13.344	0.917	0.666	4.088	0.154	–	0.566
	7.418	46.701	2.747	2.747	37.912	2.473	–	100
3 Warble	–	14.1	3.0	72.6	85.6	103.4	591.8	870.5
+	–	11.068	2.750	48.368	25.355	17.724	12.660	13.532
Whistle	–	1.620	0.345	8.340	9.833	11.878	67.984	100
4 Contact	–	46.7	11.9	24.2	44.3	54.3	12.4	193.8
tremolo	–	36.656	10.907	16.123	13.122	9.308	0.265	3.013
	–	24.097	6.140	12.000	22.859	28.019	6.398	100
5 Shriek	–	24.8	1.0	45.5	184.0	418.7	4,058.8	4,732.8
	–	19.466	0.917	30.313	54.502	71.769	86.830	73.572
	–	0.524	0.021	0.961	3.888	8.847	85.759	100
6 Squeal	–	0.5	–	–	–	–	8.8	9.3
	–	0.392	–	–	–	–	0.188	0.145
	–	5.376	–	–	–	–	94.624	100
7 Grunt	260.9	10.2	12.9	–	2.0	–	–	286.0
	57.862	8.006	11.824	–	0.592	–	–	4.446
	91.224	3.566	4.510	–	0.699	–	–	100
8 Loud call	4.5	–	7.9	–	–	–	–	12.4
	0.998	–	7.241	–	–	–	–	0.193
	36.290	–	63.710	–	–	–	–	100
9 Grunt bark	170.9	–	–	–	–	–	–	170.9
	37.902	–	–	–	–	–	–	2.657
	100	–	–	–	–	–	–	100
10 Rumble	5.5	4.9	71.4	1.0	–	–	–	82.8
	1.220	3.846	65.445	0.666	–	–	–	1.287
	6.643	5.918	86.232	1.208	–	–	–	100
Σ_{1-10}	450.9	127.4	109.1	150.1	337.6	583.4	4,674.4	6,432.9
	100	100	100	100	100	100	100	100
	7.009	1.980	1.696	2.333	5.248	9.069	72.664	100

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Table 11. Diversity of usage of the different vocal categories by *Macaca* and *Presbytis* (H_i values; $H_i max = 2.81$ for both)

A *Macaca*	*M. silenus*	*M. radiata*
Whoo	2.39	1.47
Threat rattle/growl	1.90	2.59
Non-tonal scream	1.59	2.47
Greeting	2.40	2.37
Twit/tonal scream	1.43	0.86
Shriek/squeak/squeal	0.45	0.86
Yap	1.08	1.01
Female copulation call	0.00	0.00
Male copulation call	0.00	0.00
Bark	0.80	0.00
Sneeze	0.00	0.00
Contact rattle	–	2.62
Whistle	–	1.18
Loud call	0.00	–
Babble	0.00	–
B *Presbytis*	*P. johnii*	*P. entellus*
Hiccup	2.55	2.50
Cough	2.29	1.74
Warble/whistle	1.27	1.50
Contact tremolo	1.36	2.37
Shriek	1.14	0.79
Squeal	1.87	0.30
Bark	0.76	0.54
Loud call	0.62	0.95
Grunt bark/cough bark	0.16	0.00
Ho-ho call	1.85	–
Rumble	–	0.76

Table 12. Diversity of vocal behaviour of the different age/sex classes of *Macaca* and *Presbytis* (H_i)

Age/sex class	*M. silenus*	*M. radiata*	*P. johnii*	*P. entellus*
$H_i max$:	3.59	3.70	3.32	3.32
Adult ♂	1.18	0.63	1.32	1.26
Adult ♀	0.30	1.86	1.85	2.51
Subadult ♂	0.42	1.71	1.68	1.65
Subadult ♀	0.22	1.48	2.25	1.73
Juvenile ♂	1.16	2.43	1.22	1.72
Juvenile ♀	1.19	2.37	1.08	1.19
Infant ♂+♀	1.20	2.20	0.62	0.60

where a is the number of different age/sex classes and P(i) is the calculated relative frequency with which each class uttered a given call.

Table 11 shows values for $H_i max$ and H_i obtained from the data in tables 7–10. A comparison of the two indices shows that in *M. silenus* only 2 calls were uttered by the different age/sex classes of approximately equal rates. For 3 other calls, the H_i value was close to the theoretical maximum (about 50%), whereas for the remaining 8 categories the values of H_i were well below the maximum, indicating marked disparity in their usage. By contrast, in the case of *M. radiata*, the indices of 7 calls were close to the maximum ($H_i \geq 50\%$ of $H_i max$). The comparison of the indices for the two macaques shows that the H_i values are similar only for 2 calls.

In the two langur species, the values illustrate a greater degree of symmetry in vocal behaviour. In both species, indices of 2 calls are close to the maximum, and in *P. johnii* another 7 calls have relatively high H_i indices (approx. 50% of $H_i max$). Altogether, 4 calls of the two langur species had indices with similar values.

The same method was used to calculate the diversity of vocal behaviour of the different age/sex classes as expressed by the degree to which each call/vocal category contributes to the total vocal output of a given age/sex class. In this case, the coefficient a represents the number of different vocal categories of the respective species, whereas P(i) gives the

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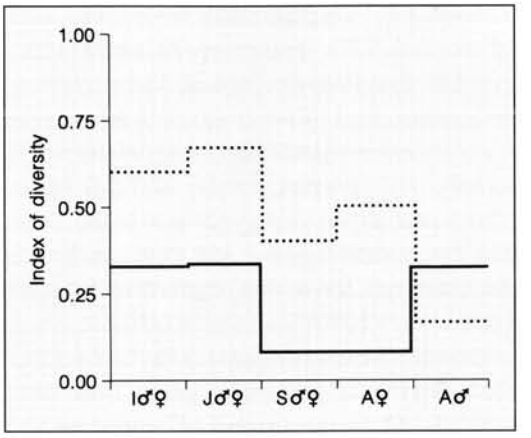


Fig. 3. Age- and sex-specific differences of vocal behaviour in *M. silenus* (solid line) and *M. radiata* (dotted line) expressed by the index of diversity DI = H_i/H_imax. The figures for the age classes I (infant), J (juvenile) and S (subadult) represent the mean values for both sexes combined as shown in table 12. A = Adult.

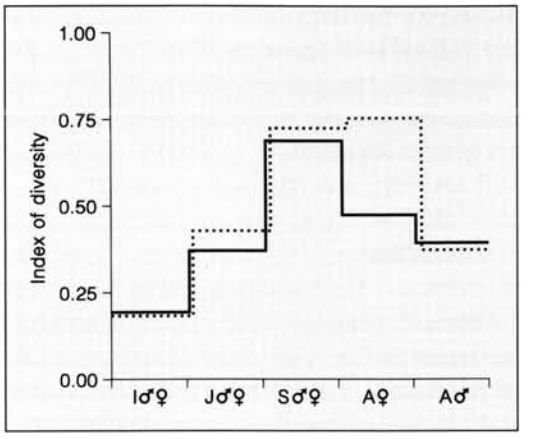


Fig. 4. Age- and sex-specific differences of vocal behaviour in *P. johnii* (solid line) and *P. entellus* (dotted line) expressed by the index of diversity DI = H_i/H_imax. The figures for the age classes I (infant), J (juvenile) and S (subadult) represent the mean values for both sexes combined as shown in table 12. A = Adult.

relative frequency with which the varius vocal categories were uttered by a given age/sex class. The results of this calculation are shown in table 12.

A comparison of the H_i values of the two macaque species reveals that, except for adult males, the diversity in vocal behaviour of *M. silenus* was consistently lower than in *M. radiata.* Marked interspecific differences between animals from equivalent age groups occurred only with adult males and females. The H_i values of the different age/sex classes of the two langurs (table 12) make it clear that, unlike *Macaca,* the diversity of vocal behaviour increases with the age of the vocalizer and reaches the highest values in adult females *(P. entellus)* and subadult females *(P. johnii).* In the two langur species, interspecific disparities within the same age class were more pronounced than in the macaques.

Figures 3 and 4 present the indices of diversity DI = H_i/H_imax) for the vocal behaviour of the four age classes.

Despite the consistent difference in diversity, the figures show a remarkably high degree of symmetry between the two macaque species: infants and juveniles account for the highest level of diversity, and the most prominent contrast occurs between juveniles and subadults. Interspecific differences exist in the relation between adult males and females; however, whereas adult males show the least diversity in *M. radiata*, adult males of *M. silenus* reach a level well above that of their female peers. A similar degree of symmetry can be seen for the two langur species (fig. 4). In the first three classes, diversity increases in regular steps. In *P. johnii*, diversity of adult females decreases to the level of juveniles, whereas adult females of *P. entel-*

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lus show a further rise in diversity, reaching the highest level found in all four species. In contrast to the pronounced inequality for female langurs, the values for the adult males are almost identical.

Discussion

The comparison of the structural and behavioural patterns of vocal communication of the four primate species indicated a general concordance within the two genera *(Macaca, Presbytis)* but pronounced differences between genera. Within each genus, the number of equivalent vocal patterns was high, and most calls were used with similar frequencies. In *Macaca,* marked differences (FI ≥ 0.5) in the use of equivalent pairs of calls occurred with 5 calls. In *Presbytis,* by contrast, only 3 pairs of equivalent calls showed marked asymmetries.

In *Macaca,* the greatest disparity was found in 2 noisy sounds (scream and growl), both uttered more frequently by *M. radiata* than by *M. silenus.* As the emission of these calls was closely related to agonistic intragroup encounters in both species, the differences in vocal behaviour suggest a comparatively higher rate of such interactions in *M. radiata.* In the field, bonnet macaques form cohesive groups with interindividual distances of 0.5–1.5 m [28]. Spatial distribution within groups of *M. silenus* is characterized by relatively cohesive female clans keeping interindividual distances of approximately 5 m [29]. Between different clans, however, distances may be 600 m or more [19, pers. observation]. The recorded interindividual distances in *M. radiata* groups may lead to a comparatively high level of social conflict and, consequently, to more agonistic interactions. During data collection with *M. silenus,* 0.07 intergroup encounters/day (n = 177 days) were recorded, and according to another study [30] only 11% of the encounters are accompanied by agonistic interactions. By contrast, in *M. radiata,* intergroup encounters occurred at a higher rate (0.5 encounter/day, n = 158 days) and were characterized by severe aggressive interactions.

Another obvious disparity in the usage of equivalent vocal patterns occurred with contact calls. The large number of contact whoos uttered by *M. silenus* may be essential for spatial regulation among the less cohesive groups in the dense rain forest habitat. However, the pronounced age differences in whoo call vocalization in this species deserve further examination. As mentioned above, groups of *M. silenus* consist of well-defined female clans. Clear calls of *Macaca arctoides* and *Macaca fuscata* are known to possess individual-specific structural features [31, 32], and in the field it was possible to identify most of the mature individuals of *M. silenus* by their contact whoos alone. Provided that a contact whoo signals not only the location but also the identity of the vocalizer, this vocalization may serve to increase cohesion between the members of the same clan and, at the same time, maintain sufficient distances between the different clans. For the time being, the most likely explanation for the low number of contact whoos in *M. radiata* seems to be the absence of well-defined female clans and the close spatial proximity among group members.

Considering the inequality in rates of long-distance whoos, two factors may be relevant. In *M. silenus,* the frequent utterance of contact whoos provides an almost continuous acoustic indicator of the current posi-

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tion and spatial distribution of the various clans. Thus, loss of contact between members of the group may occur less frequently than in *M. radiata*. After group fission (the common stimulus for long-distance whoo emission in *M. radiata*), the male leader of an *M. silenus* group gives long-distance calls [23]. These calls carry as far as 1 km and lead to distance reduction between the vocalizer and scattered subgroups [5, 20].

The different rates of mating vocalizations of females remain to be explained. Studying mating behaviour of female *Macaca fascicularis*, Deputte and Goustard [33] reasoned that the 'utterance of female copulatory vocalizations is essentially due to the female's own copulatory behaviour...'. Observations of captive lion-tailed macaques suggest that female copulation calls may be directed towards other mature female group members rather than to the consorting male [20]. The high levels of mating-related, intrasexual competition among females in the wild [34] support this assumption. When we take into account the (a) synchronization of cycling among female group mates, (b) the restricted number of mature males living in each group and (c) the time-consuming mode of mating (multiple mounting), the reproductive success of a female may depend primarily on her ability to prevent competing females from interfering with her efforts to establish a consort relation with a male. In bonnet macaques mating is also restricted to a particular season. However, in this species females are highly promiscuous [35], and males are single mounters [36], achieving ejaculation at the end of one intromissive mount in 91% of their copulations [37]. Observations from the laboratory and field indicate that females may prefer mating partners ranking well below the dominant group leaders [38]. The larger number of 'silent copulations' in this species may therefore be due to both the frequent mating with low-ranking males and the reduced competition among adult females.

For the two langur species, different frequencies in the use of equivalent calls were less pronounced than in the two macaque species. *P. johnii* produced the higher rates of alarm calls and hiccups, while *P. entellus* was more active in the case of the two-phase barks (cough bark, grunt bark). In both species, adult females and immatures of both sexes gave alarm calls in response to the appearance of a potential predator. In contrast, hiccups were uttered by males and females of all age classes and were heard not only after detection of a potential predator but also in the context of group encounters. Considering the number and predator category (leopard) the samples of alarm calls of both species can be regarded as equivalent, and the different rates of this vocalization are puzzling. Tables 9 and 10 show that in the case of hiccups the corresponding age/sex classes of both species contributed with almost identical rates. Hence, considering the differences in group size (table 1) the rate should be higher in *P. entellus* than in the other species, but the opposite is true. The large number of hiccups uttered by *P. johnii* may be partly due to the variety of stimuli evoking this vocalization, which include not only terrestrial predators and other animals but even objects such as large pieces of driftwood. Moreover, besides signalling the detection of a predator or strange object, adult males of *P. johnii* regularly gave hiccups as part of their loud-call displays. In *P. entellus*, hiccups were heard only in response to the appearance of a neighbouring group, and still in these situations grunt barks were

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more common than hiccups. As in the case of other calls (e.g. harsh bark), the use of hiccups by *P. johnii* therefore seems to be less context specific than in *P. entellus.* Moreover, field observations indicate that the grunt bark of *P. entellus* may serve a dual function [26] and thus correspond to both the cough bark and the hiccup of *P. johnii.* Therefore, the different rates of the two vocal patterns (hiccup, grunt bark) may be the result of species-specific differences in their communicative function.

Striking dissimilarities between *Macaca* and *Presbytis* were found in age- and sex-specific vocal behaviour, as expressed by the index of diversity. Within each species group, however, interclass differences in diversity were relatively similar. Even in the case of *Macaca,* where the real values of diversity differed markedly, the patterns of variation across corresponding age/sex groups showed a high degree of symmetry between the two species.

The ontogeny of social behaviour of immature primates is strongly affected by two factors: interactions between the infant and group members other than the mother [39], and the quality of maternal care [40, 41]. During the first weeks of life, infants of most macaque species stay almost exclusively with their mothers. Contacts between the infant and other group members are – dependent on the species, the mother's rank and the age of the infant – regulated strictly by the mother [42]. By contrast, in various langur species it is known that infant transfers occur during the very first day of life [43–45] and that the mother's influence on (or interest in) the control of such transfers is relatively limited [43, 45]. The data so far available do not permit a systemic evaluation of the effect of maternal care on the ontogeny of social communication. Nevertheless, because of the close relationship between vocalization and emotional state of the vocalizer [46], on the one hand, and proximity between vocalizer and potential recipient [47] and the quality of social interaction [48, 49], on the other, the social environment to which immatures are exposed is certainly of exceptional importance for the ontogenetic development of vocal communication. Therefore, it is conceivable that some of the intergeneric differences in vocal behaviour between *Macaca* and *Presbytis* may be related to differences in maternal behaviour and treatment of infants by other group members.

In studies dealing with the ontogenetic development and socialization of non-human primates, various changes in form and function of communicative behaviour have been reported [5, 13, 50–52]. One aspect concerns the relationship between visual and vocal signals. In many primate species, most vocal patterns seem to be present soon after birth [53–55]. By contrast, many visual patterns are related primarily to functions such as nourishment, locomotion or protection and acquire their communicative function secondarily during the process of maturation [13, 39]. Studies of various macaque species have shown that, with increasing age, visual patterns may not only complement vocal signals but may even replace vocal cues [39, 50]. These and other reports [13] suggest that the different modes of communication may develop in different ways during ontogeny. In addition to intraspecific variation in maturation and use of communication systems, species of different taxa may differ in the size and complexity of the repertoires of vocal and visual signals. The few comparative data on visual communication in langurs and macaques so far available indicate subtle differ-

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ences in the development of this communication system [51], and several authors have reached the conclusion that, in langurs, facial expressions are more rigid and less complex than in cercopithecines [44, 56].

Previous studies investigating age and sex specificity of vocal behaviour have dealt with chimpanzees and gorillas [15] and African guenons [8]. A comparison of the data collected in these studies with our results shows a high degree of similarity between *Cercopithecus* and the two leaf monkeys *P. johnii* and *P. entellus*. In both the guenons and the langurs diversity increases with age and is lower in adult males than in adult females. Thus, we may conclude that in all of them age- and sex-specific differences in vocal behaviour are roughly the same. On the other hand, the diversity profiles of the two macaque species differ from the figures obtained from chimpanzees and gorillas as well as from those of guenons and langurs. When comparing only the indices (H_i values) for the infants of the two apes and the two macaque species, however, striking similarities become apparent. Here, the diversity index of chimpanzees (2.10) is very similar to that of bonnet macaques (2.20), and that of infant gorillas (1.18) is almost identical to that of lion-tailed macaques (1.20). More interesting than this statistical correspondence is the high degree of symmetry in interspecific differences (*P. troglodytes* vs. *G. gorilla*, *M. radiata* vs. *M. silenus*) which persists even in juvenile animals of the four species.

Based on the analyses here, it is concluded that: (1) behavioural and structural aspects of vocal communication are characterized by a high degree of intrageneric similarity independent of the environmental adaptation of the respective species; (2) differences in the rate of vocalization of equivalent calls are not simply the result of disparities in size and composition of the group but may reflect (a) qualitative differences in related behavioural interactions and (b) variation of the function specificity of particular calls; (3) the general asymmetries of vocal behaviour between *Macaca* and *Presbytis* may be related to different modes of infant care as well as differences in communication by visual signals; (4) comparisons of the diversity of vocal behaviour of the macaques and langurs with that in other species suggest a high concordance between leaf monkeys and guenons and some parallels between macaques and the great apes.

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References

1 Smith HJ, Newman JD, Symmes D: Vocal concomitants of affiliative behavior in squirrel monkeys; in Snowdon CT, Brown CH, Petersen MR (eds): Primate Communication. New York, Cambridge University Press, 1982, pp 30–49.

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128.111.121.42 - 5/5/2016 12:29:44 PM

2 Jürgens U: A neuroethological approach to the classification of vocalization in the squirrel monkey; in Snowdon CT, Brown CH, Petersen MR (eds): Primate Communication. Cambridge, Cambridge University Press, 1982, pp 50–62.
3 Geissmann T: Female capped gibbon *(Hylobates pileatus)* sings male song. J Human Evol 1983;12: 667–671.
4 Dunaif J: Sexual dimorphism of vocal apparatus and vocal behavior in non-human primates. Am J Phys Anthropol 1980;52:222.
5 Green S: Sex differences and age gradations in vocalizations of Japanese and lion-tailed monkeys (*Macaca fuscata* and *Macaca silenus*). Am Zool 1981;21:165–183.
6 Haimoff E: A possible case of deception in the song of the siamang. Am J Primatol 1984;6:407.
7 Fossey D: Vocalizations of the mountain gorilla *(Gorilla gorilla beringei)*. Anim Behav 1972;20: 36–53.
8 Gautier JP, Gautier-Hion A: Vocal communication within a group of monkeys: An analysis by biotelemetry; in Snowdon CT, Brown CH, Petersen MR (eds): Primate Communication. Cambridge, Cambridge University Press, 1982, pp 5–29.
9 Marler P: Social organization, communication and graded signals: the chimpanzee and the gorilla; in Bateson, PP, Hinde RA (eds): Growing Points in Ethology. London, Cambridge University Press, 1976, pp 239–279.
10 Wiley RH: The evolution of communication: Information and manipulation; in Halliday TR, Slater PJ (eds): Animal Behaviour. Oxford, Blackwell Scientific Publications, 1983, vol 2, pp 156–189.
11 Vogel C: Acoustical communication among free-ranging common Indian langurs *(Presbytis entellus)* in two different habitats of North India. Am J Phys Anthropol 1973;38:469–480.
12 Eisenberg JF: Communication Mechanisms and Social Integration in the Black Spider Monkey, *Ateles fusciceps robustus*, and Related Species. Washington, Smithsonian Institution Press, 1976.
13 Gautier JP, Gautier A. Communication in Old World monkeys; in Sebeok TA (ed): How Animals Communicate. Bloomington, Indiana University Press, 1977, pp 890–964.
14 Snowdon CT: Vocal communication; in Mitchel AG, Erwin J (eds): Comparative Primate Biology. New York, Liss, 1986, vol 2, pp 175–196.
15 Marler P, Tenaza R: Signalling behavior of apes with special reference to vocalization, in Sebeok TA (ed): How Animals Communicate. Bloomington, Indiana University Press, 1977, pp 965–1033.
16 Harcourt AH, Stewart KJ, Harcourt DE: Vocalizations and social relationships of wild gorillas: A preliminary analysis; in Taub DM, King FA (eds): Current Perspectives in Primate Social Dynamics. New York, Van Nostrand Reinhold, 1986, pp 346–356.
17 Poirier FE: The Nilgiri langur *(Presbytis johnii)* troop: Its composition, structure, function and change. Folia Primatol 1969;10:20–47.
18 Kumar A: The Ecology and Population Dynamics of the Lion-Tailed Macaque *(Macaca silenus)* in South India; PhD thesis University of Cambridge, 1987.
19 Roonwal ML, Mohnot SM: Primates of South Asia. Cambridge, Harvard University Press, 1977.
20 Hohmann GM, Herzog MO: Vocal communication in lion-tailed macaques *(Macaca silenus)*. Folia Primatol 1985;45:148–178.
21 Poirier F: The communication matrix of the Nilgiri langur *(Presbytis johnii)* of South India. Folia Primatol 1970;13:92–136.
22 Horwich R: The whooping display in Nilgiri langurs: An example of daily fluctuations superimposed on a serial trend. Primates 1976;17:419–431.
23 Herzog MO, Hohmann GM: Male loud calls in *Macaca silenus* and *Presbytis johnii:* A comparison. Folia Primatol 1984;43:189–197.
24 Hohmann G: Vocal communication of wild bonnet macaques *(Macaca radiata)*. Primates 1989; 30:325–345.
25 Hohmann G: Comparative study of vocal communication in two Asian leaf monkeys, *Presbytis johnii* and *P. entellus*. Folia Primatol 1989;52:27–57.
26 Hohmann G: Akustische Kommunikationssysteme catarrhiner Primaten: Vergleichende Untersuchungen am Modell der simischen Primatenfauna Südindiens; PhD thesis Universität München, 1988.

Downloaded by:
Univ. of California Santa Barbara
128.111.121.42 - 5/5/2016 12:29:44 PM

27 Shannon CE, Weaver W: The Mathematical Theory of Communication. Urbana, University of Illinois Press, 1949.
28 Ali R: The Ecology and Social Behaviour of the Agastyamalai Bonnet Macaque *(Macaca radiata diluta);* thesis University of Bristol, 1981.
29 Green S: Feeding, spacing and movements as correlates of troop size in lion-tailed macaque; in Chivers DJ, Herbert J (eds): Recent Advances in Primatology. London, Academic Press, 1978, vol 1, pp 343–345.
30 Kumar A, Kurup GU: Inter-troop interactions in the lion-tailed macaque, *Macaca silenus*; in Heltne PG (ed): The Liontailed Macaque: Status and Conservation. New York, Liss, 1985, pp 91–108.
31 Lillehei RA, Snowdon CT: Individual and situational differences in the vocalizations of young stumptail macaques *(Macaca arctoides).* Behaviour 1978;65:270–281.
32 Mitani M: Voiceprint identification and its application to sociological studies of wild Japanese monkeys *(Macaca fuscata yakui).* Primates 1986; 27:397–412.
33 Deputte BL, Goustard M: Copulatory vocalizations of female macaques *(Macaca fascicularis):* Variability factor analysis. Primates 1980;21:83–99.
34 Kumar A: Sexual harassment in *Macaca silenus.* Abstr 12th Congr IPS, Brasilia, 1988.
35 Simonds PE: The bonnet macaque in South India; in DeVore I (ed): Primate Behaviour: Field Studies of Monkeys and Apes. New York, Holt, Rinehart & Winston, 1965, pp 175–196.
36 Caldecott JO: Mating patterns, societies and the ecogeography of macaques. Anim Behav 1986;34: 208–220.
37 Shiveley C, Clarke S, King N, et al: Patterns of sexual behavior in male macaques. Am J Primatol 1982;2:373–384.
38 Beckerman-Glick B: Ontogenetic and psychobiological aspects of the mating activities of male *Macaca radiata*; in Lindburg DG (ed): The Macaques: Studies in Ecology, Behavior and Evolution. New York, Van Nostrand Reinhold, 1980, pp 345–369.
39 Chevalier-Skolnikoff S: The Ontogeny of Communication in the Stumptail Macaque *(Macaca arctoides).* Contrib Primatol. Basel, Karger, 1974. vol 2.
40 Hinde RA: Development of social behavior; in Schrier AM, Stollnitz F (eds): Behavior of Nonhuman Primates. New York, Academic Press, 1971, vol 3, pp 1–68.
41 Rosenblum LA, Kaufmann IC: Laboratory observations of early mother-infant relations in pigtail and bonnet macaques; in Altman SA (ed): Social Communication among Primates. Chicago, University of Chicago Press, 1967, pp 33–41.
42 McKenna JJ: The evolution of allomothering behavior among colobine monkeys: Function and opportunism in evolution. Am Anthropol 1979; 81:818–840.
43 Jay P: The common langur of North India; in DeVore I (ed): Primate Behavior: Field Studies of Monkeys and Apes. New York, Holt Rinehart & Winston, 1965, pp 197–249.
44 Poirier FE: The Nilgiri langur *(Presbytis johnii)* of South India; in Rosenblum LA (ed): Primate Behavior: Developments in Field and Laboratory Research. New York, Academic Press, 1970, vol 4, pp 251–383.
45 Bennett EL: The occurrence of twins and accompanying behavioural changes in the banded langur, *Presbytis melalophos.* Primates 1988;29:557–564.
46 Jürgens U: Vocalizations as an emotional indicator: A neuroethological study on squirrel monkeys. Behaviour 1979;69:88–117.
47 Robinson JG: Vocal systems regulating within-group spacing; in Snowdon CT, Brown CH, Petersen MR (eds): Primate Communication. Cambridge, Cambridge University Press, 1982, pp 94–116.
48 Klotz M: Zum Problem der bidirektionalen akustischen Kommunikation bei der Gattung *Macaca.* Z Psychol 1973;180/181:283–310.
49 Goedeking P: Zur Kovariation von Lautstrukturvariabilitäten und sozialem Spiel bei Lisztaffen *(Saguinus oedipus oedipus):* Eine quantitative Untersuchung mit Hilfe rechnergesteuerter Lautanalysemethoden; PhD thesis Universität Bielefeld, 1985.
50 Angst, W: Das Ausdrucksverhalten des Javaneraffen, *Macaca fascicularis* Raffles. Berlin, Parey, 1974.
51 Chevalier-Skolnikoff S: A cognitive analysis of facial behavior in Old World monkeys, apes and human beings; in Snowdon CT, Brown CH, Petersen MR (eds): Primate Communication. Cam-

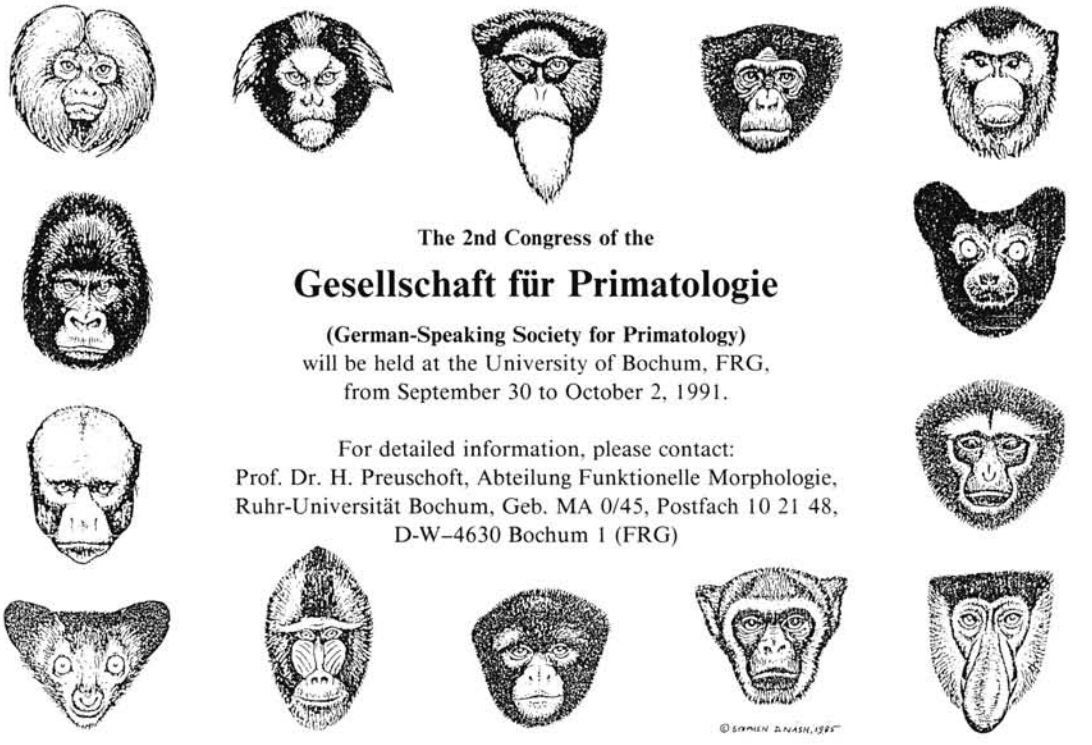
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bridge, Cambridge University Press, 1982; pp 303–368.
52 Seyfarth RM, Cheney DL: The ontogeny of vervet monkey alarm calling behavior: A preliminary report. Z Tierpsychol 1980;54:37–56.
53 Gautier JP: Field and laboratory studies of the vocalizations of talapoin monkeys *(Miopithecus talapoin):* structure, function, ontogenesis. Behaviour 1974;49:1–64.
54 Bertrand M: The behavioral repertoire of the stumptail macaque; in Hofer H, Schultz A, Starck D (eds): Bibliotheca Primatologica. Basel, Karger, 1969, No. 11.
55 Winter P, Handley P, Ploog D, Schott D: Ontogeny of squirrel monkey calls under normal conditions and under acoustic isolation. Behaviour 1973;47:230–239.
56 Vogel C: Ökologie, Lebensweise und Sozialverhalten der grauen Languren in verschiedenen Biotopen Indiens. Berlin, Parey, 1976.

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G. Hohmann
Max-Planck-Institut für Psychiatrie
D-W–8000 München 40 (FRG)

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